Smart Pointers and Ownership

Who owns an object: `unique_ptr`, `shared_ptr` and `weak_ptr`, the control block, cycles, and how to pass ownership around. Builds on RAII and Move Semantics.

Who owns it? The three smart pointers

RAII wrappers around `new` / `delete` (`<memory>`) that make **ownership explicit in the type**. The owner is whoever must free the object.

TYPE	OWNERSHIP	COPYABLE?	COST
`std::unique_ptr<T>`	exclusive: one owner	**no**, only movable	same as a raw pointer
`std::shared_ptr<T>`	shared: reference counted, freed when the last owner goes	yes (bumps the count)	extra control block, atomic count
`std::weak_ptr<T>`	none: observes a `shared_ptr` without keeping it alive	yes	small
raw pointer / reference	none: "I use it, I do not own it"	yes	none

> **DEFAULT** A plain value member or local first. Then `unique_ptr` when it must live on the heap (polymorphism, large, optional). `shared_ptr` only for truly shared ownership.

A FUNCTION THAT…	TAKES
only uses the object	`const Widget&` (or `Widget*` if it may be null): never the smart pointer
takes over ownership	`std::unique_ptr<Widget>` by value: the caller must `std::move`
keeps a share	`std::shared_ptr<Widget>` by value
gives ownership to the caller	returns `std::unique_ptr<Widget>` (a factory)

```
std::unique_ptr<Widget> makeWidget(std::string n);        // factory: sole ownership to the caller
void use(const Widget& w);                                // just uses: pass the object, not the owner
void take(std::unique_ptr<Widget> w);                     // takes ownership

auto w = makeWidget("alpha");
use(*w);
take(std::move(w));                                       // w is now null
```

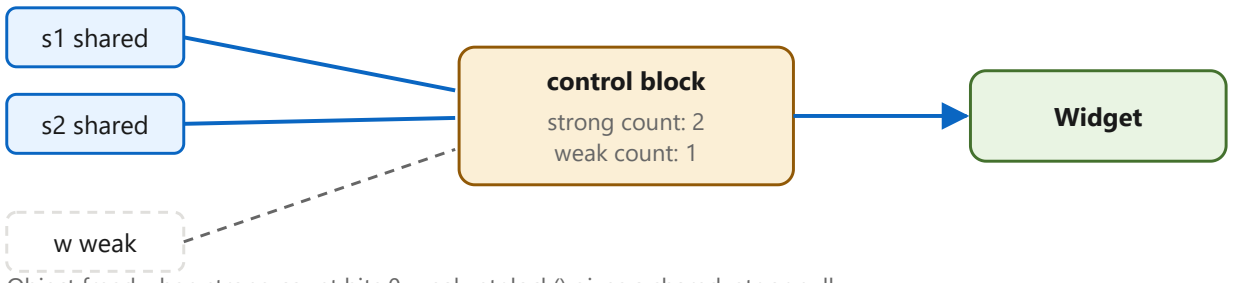
`unique_ptr` in detail

```
auto p = std::make_unique<Widget>(1, "a");    // preferred over `new`
// auto q = p;                                // ERROR: unique_ptr cannot be copied
auto q = std::move(p);                        // OK: ownership moves, p is now null
```

- **Zero overhead:** with the default deleter `sizeof(unique_ptr<T>) == sizeof(T*)` (checked). A function-pointer deleter makes it twice as big; a stateless lambda deleter adds nothing.
- **Custom deleter** for anything that is not `delete`: `std::unique_ptr<std::FILE, decltype(closer)> f(std::fopen( ... ), closer);` closes the file on scope exit.
- **Arrays:** `std::make_unique<int[]>(3)` uses `delete[]`. Prefer `std::vector` unless you need a fixed block.
- **Polymorphism:** `unique_ptr<Base>` holding a `Derived` needs a **virtual destructor** on `Base`, as with any `delete` through a base pointer.
- `.get()` returns the raw pointer without giving up ownership; `.release()` gives it up (you must delete it); `.reset()` destroys the object now.

KEY POINT Why `make_unique`? No naked `new`, the type is written once, and before C++17 `f(unique_ptr<A>(new A), g())` could leak when `g()` threw between the two steps.

`shared_ptr` and the control block



Object freed when strong count hits 0. weak_ptr.lock() gives a shared_ptr or null.

unique_ptr has no control block: zero overhead, one owner.

```
auto s1 = std::make_shared<Widget>(2, "b");  // one allocation for object + control block
auto s2 = s1;                                // two owners, use_count() == 2
```

- A `shared_ptr` is **two pointers**: to the object and to a **control block** holding the strong count, the weak count and the deleter.
- `make_shared` allocates object and control block together (one allocation). `shared_ptr<T>(new T)` allocates twice.
- The deleter is not part of the `shared_ptr` type (unlike `unique_ptr`): type-erased in the control block.
- The counts are **atomic**: copying or destroying `shared_ptr`s on several threads is safe. The **object** is not protected, and changing the same `shared_ptr` variable from two threads is a race.
- Copying costs an atomic increment. On hot paths pass `const shared_ptr&` or the object itself, not a copy per call.

Reference cycles and `weak_ptr`

If `A` holds a `shared_ptr<B>` and `B` holds a `shared_ptr<A>`, each keeps the other alive: **neither is ever destroyed**. Compiled: after the scope, both objects were still alive (leaked). Make the back link (child to parent) a `weak_ptr`: both were destroyed.

```
struct Node { std::shared_ptr<Owner> parent; };        // cycle: leak
struct WNode { std::weak_ptr<OwnerW> parent; };        // weak back-pointer: freed

std::weak_ptr<Widget> w = s1;                          // observe only: does not change use_count
if (auto sp = w.lock()) { /* still alive: use sp for as long as you hold it */ }   // lock() gives a shared_ptr or null
w.expired();                                           // true once every owner is gone
```

- `lock()` is the only way to use the object: it gives a `shared_ptr` that keeps it alive while you work, or null.
- Typical uses: parent links, caches, observer lists that must not keep subscribers alive.
- A `weak_ptr` keeps the **control block** alive (not the object) until the last weak reference goes.

Pitfalls and `enable_shared_from_this`

ONE CONTROL BLOCK PER OBJECT

```
auto raw = new Widget;
std::shared_ptr<Widget> a(raw);
std::shared_ptr<Widget> b(raw);        // a second, independent control
block: double delete!
```

Never build a second `shared_ptr` from a raw pointer or from `this`. Copy an existing one.

When a member function needs a `shared_ptr` to its own object (to register itself, to hand to a callback), inherit `std::enable_shared_from_this` and call `shared_from_this()`:

```
struct Session : std::enable_shared_from_this<Session> {
    std::shared_ptr<Session> self() { return shared_from_this(); }   // a
safe second owner, SAME control block
};
auto s = std::make_shared<Session>();
auto t = s->self();                    // use_count() == 2
Session raw; raw.self();               // throws std::bad_weak_ptr: no
shared_ptr owns it (compiled)
```

PITFALL	FIX
`shared_ptr` cycle	`weak_ptr` for the back link
`shared_ptr` from `this` or a raw pointer	`enable_shared_from_this`
`shared_ptr` by default "to be safe"	`unique_ptr`: you can always convert to shared later
storing `.get()` somewhere that outlives the owner	keep the smart pointer, or a `weak_ptr`
`delete` on a `.get()` result	never: the smart pointer will delete it again

Further reading by section

Where each section of this lesson is explained in more depth. Page numbers are PDF viewer pages.

Who owns it? The three smart pointers

C++ Primer (Lippman, Lajoie, Moo)

- 12.1 Dynamic memory and smart pointers (PDF pages 479 to 504)

Programming: Principles and Practice (Stroustrup)

- 18.4 to 18.5 Resources and exceptions; resource-management pointers (RAII, unique_ptr) (PDF pages 542 to 553)

C++ Move Semantics (Josuttis)

- Chapter 13 Move-only types (PDF pages 213 to 218)

unique_ptr in detail

C++ Primer (Lippman, Lajoie, Moo)

- 12.1 Dynamic memory and smart pointers (PDF pages 479 to 504)

Effective Modern C++ (Meyers)

- Item 18, Use std::unique_ptr for exclusive-ownership resource management (PDF pages 136 to 143)
- Item 21, Prefer make_unique and make_shared to new (PDF pages 157 to 165)

shared_ptr and the control block

C++ Primer (Lippman, Lajoie, Moo)

- 12.1 Dynamic memory and smart pointers (PDF pages 479 to 504)

Effective Modern C++ (Meyers)

- Item 19, Use std::shared_ptr for shared-ownership resource management (PDF pages 143 to 152)

Reference cycles and weak_ptr

C++ Primer (Lippman, Lajoie, Moo)

- 12.1 Dynamic memory and smart pointers (PDF pages 479 to 504)

Effective Modern C++ (Meyers)

- Item 20, Use std::weak_ptr for shared_ptr-like pointers that can dangle (PDF pages 152 to 157)